

Ethical Dilemmas in Genetic Testing

Current 26-page faculty deck primary - official Week 4 HPC objectives integrated - CGD genetic-testing case, informed consent, privacy, family implications, discrimination, DNA storage, and first-step ethics reasoning

2 official objectives

HPC

Informed consent

Family implications

Privacy

First-step reasoning

SOURCE PRIORITY: the uploaded current faculty deck controls terminology, examples, and emphasis. The official Week 4 learning-objective sheet defines what must be mastered. This companion reorganizes the source material for retrieval and application without replacing the deck.

Official Week 4 Objectives

1. Identify ethical issues in genetic testing that make informed consent more complex than in many other laboratory tests.
2. Given a complex clinical scenario, choose the most appropriate first step in addressing a major ethical dilemma related to genetic testing.

Deck pp. 4-12

Clinical anchor: suspected chronic granulomatous disease shows why a molecular diagnosis matters

Context

The case follows a febrile 4-week-old boy with a maternal family history of affected males, Aspergillus pulmonary infection, and an absent dihydrorhodamine respiratory burst. The working diagnosis is **chronic granulomatous disease (CGD)**, but the lecture asks whether a clinical diagnosis alone is enough.

Why genetic testing adds value	Lecture application
Precise diagnosis	CGD can result from several genes with different inheritance patterns, including an X-linked form and autosomal forms.
Treatment planning	A molecular diagnosis may matter for gene-targeted or gene-based therapies.
Family counseling	Results can clarify carrier risk, reproductive counseling, and which relatives may be at risk.
Predictive / proactive medicine	Genomics can support diagnosis before symptoms, prognosis, pharmacogenomics, cancer genetics, and targeted therapy.

Ethics begins because the result does not belong only to the disease in front of you. A germline result can reveal information about the patient's future and about biologic relatives who never consented to testing.

Deck pp. 12-15

Why genetic testing makes consent unusually complex

LO 1

Durable information

DNA-based information can remain relevant throughout life and may be reinterpreted as knowledge changes.

Family implications

A result about one person can imply carrier status, disease risk, or reproductive risk in relatives.

Predictive burden

Testing can reveal risk before symptoms and can create psychological consequences even when penetrance or severity is uncertain.

Secondary use

Stored DNA or results can raise questions about future testing, research, commercial use, and third-party access.

Social consequences

The lecture highlights concerns about discrimination, insurance, employment, family planning, and spirituality.

Equity

Cost, insurance coverage, access to accredited testing, and commercialization can influence who benefits from genomic medicine.

Why informed consent is more than permission for a blood draw: the patient may be consenting to information that is predictive, familial, durable, and potentially sensitive beyond the immediate clinical question.

Deck p. 13

Informed consent is the central starting point in the lecture

LOs 1-2

Deck rule: informed consent that addresses benefits and risks is an absolute prerequisite for clinical or research genetic testing. The usual decision-maker is the patient or an authorized parent / legal guardian.

- Exceptions are uncommon and depend on applicable law, institutional policy, and the specific circumstances.
- If an exception is considered, it should be addressed **before** sample collection and testing.
- The deck treats obtaining and processing a specimen without consent as ethically unacceptable because it conflicts with autonomy, confidentiality, beneficence, and nonmaleficence.
- If a consent violation has already occurred, the deck identifies the **institutional ethics committee** as the starting point.

Exam trap: a potentially beneficial result does not automatically justify bypassing consent. First establish who can authorize the test and whether valid consent exists.

Deck pp. 15-23

The ethical domains span medical, legal, psychological, social, and spiritual concerns

LO 1

Domain	Questions emphasized in the lecture
Availability / commercialization	Who can afford testing? Is the laboratory clinically accredited? Who benefits financially from samples or discoveries?
Psychological impact	Would the patient want to know about a severe future disease before symptoms? How do variable penetrance and uncertain expression change that choice?
Spirituality / values	How can beliefs about destiny, suffering, pregnancy, or intervention shape decision-making?
Discrimination	Could genetic information affect education, loans, insurance, or employment?
Reproduction	How should carrier information, prenatal implications, or reproductive risk be handled?
DNA storage / secondary use	Who can access stored DNA, order future testing, use it in research, or use it commercially?

Family communication

Who should tell relatives? What if the patient refuses? What rights do children and other family members have to know or not know?

The same fact can create competing duties. Protecting patient confidentiality may conflict with a desire to warn relatives; respecting autonomy may conflict with a clinician's belief that testing would be medically beneficial.

Deck pp. 19-22

Privacy, discrimination, and stored DNA require separate attention

LO 1

GINA as presented in the lecture

The deck highlights the **Genetic Information Nondiscrimination Act of 2008** as protecting against use of genetic information in **health insurance and employment** decisions. For studying this lecture, preserve that scope rather than assuming all forms of insurance or all contexts are covered.

Stored DNA is not ethically neutral

After the original test, questions remain about access to results, additional testing, legal or genealogic use, commercial development, and whether the original consent covered those later uses.

Do not equate privacy with secrecy. The test may legitimately require communication among clinicians and the patient, but disclosure outside the authorized circle requires a separate justification.

Deck pp. 10, 21, 23

Genetic information can create obligations and tensions within a family

LOs 1-2

Carrier detection

A molecular result in the proband can identify relatives who may benefit from counseling or targeted testing.

Right to know / not know

Predictive information may be valuable to one person and unwanted by another.

Children

Questions include who consents, when the child should participate, privacy from relatives, and whether testing has an immediate medical benefit.

Refusal

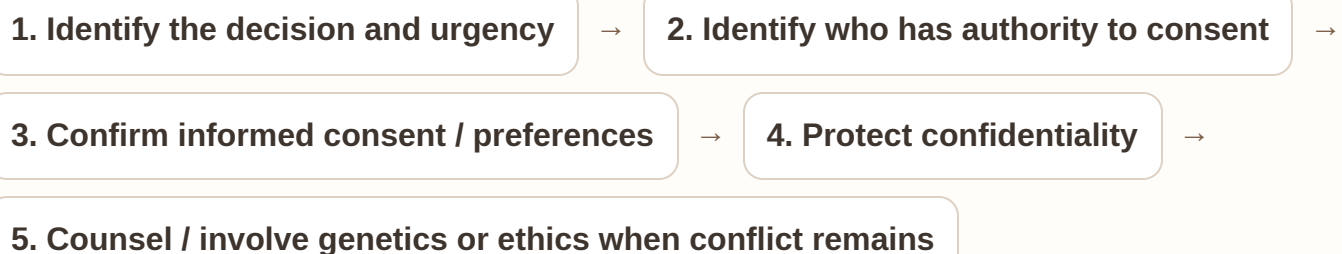
A relative's refusal to be tested does not erase the ethical problem; it changes the problem to counseling, confidentiality, and risk communication.

Patient result, family relevance: genetic information is relational, but the existence of family risk does not automatically transfer decision-making authority away from the patient.

Clinical application

A practical first-step framework for ethical genetic-testing vignettes

LO 2



Scenario cue

Most defensible first move from the lecture's framework

Testing has not yet been performed and consent is unclear	Do not test yet. Clarify decision-making authority and obtain informed consent.
Specimen was tested without appropriate consent	Stop further unauthorized use/disclosure and begin with the institutional ethics process identified in the deck.
Result implies risk to relatives	Counsel the patient about familial implications and communication while preserving confidentiality; escalate through genetics/ethics when duties conflict.
Patient fears discrimination	Clarify what the test can reveal, discuss privacy and the protections/limits presented in the course, and ensure the patient can make an informed choice.
Family demands a patient's result	Do not automatically disclose. First assess authorization, confidentiality, and the patient's preferences.

Vignette strategy: when several later actions seem reasonable, choose the action that establishes ethically valid decision-making first - usually consent, capacity/authority, confidentiality, or ethics consultation - before testing, disclosure, or contacting relatives.

Exam traps

Protect the first-step logic

Both LOs

Trap 1: "It will help the patient" is not a substitute for informed consent.

Trap 2: family benefit does not automatically authorize disclosure of the patient's result.

Trap 3: a research laboratory doing a favor is not automatically equivalent to a clinically accredited diagnostic laboratory.

Trap 4: predictive testing can have psychological and reproductive consequences even when the patient is asymptomatic.

Trap 5: GINA in the lecture is about health insurance and employment; do not mentally expand it to every possible form of insurance or social consequence.

Trap 6: stored DNA creates new consent and privacy questions beyond the original test.

30-SECOND FINISH

Genetic testing is special because the information is durable, predictive, familial, and sensitive. Consent comes first. Clarify who can authorize testing, explain benefits and risks, protect confidentiality, and use genetics/ethics resources when family duties, disclosure, or prior consent violations create conflict.